

- ```
CREATE TABLE STUDENT (
 Roll_No CHAR(6) PRIMARY KEY,
 StudentName VARCHAR(20),
 Course VARCHAR(10),
 DOB DATE
);
```
- ```
CREATE TABLE SOCIETY (
    SocID        CHAR(6) PRIMARY KEY,
    SocName      VARCHAR(20),
    MentorName   VARCHAR(15),
    TotalSeats   INT UNSIGNED
);
```
- ```
CREATE TABLE ENROLLMENT (
 Roll_No CHAR(6),
 SID CHAR(6),
 DateOfEnrollment DATE,
 PRIMARY KEY (Roll_No, SID),
 FOREIGN KEY (Roll_No) REFERENCES STUDENT(Roll_No),
 FOREIGN KEY (SID) REFERENCES SOCIETY(SocID)
);
```
- ```
INSERT INTO STUDENT VALUES
('S00001', 'Aarav Kumar', 'computer s', '2001-05-15'),
('S00002', 'Aditi Sharma', 'chemistry', '2002-08-20'),
('S00003', 'Rahul Verma', 'physics', '2000-11-10'),
('X00009', 'Sneha Gupta', 'computer s', '2001-02-25'),
('Z00009', 'Vikram Singh', 'maths', '1999-07-30'),
('S00006', 'Priya Das', 'chemistry', '2003-01-12'),
('S00007', 'Amit Patel', 'english', '2001-09-05'),
('S00008', 'Neha Jain', 'history', '2002-04-18');
```
- ```
INSERT INTO SOCIETY VALUES
('s1', 'NSS', 'Mr. A Gupta', 50),
('s2', 'Debating', 'Ms. R Kaur', 30),
('s3', 'Dancing', 'Mr. S Sharma', 40),
('s4', 'Sashakt', 'Dr. V Gupta', 25),
('s5', 'Music', 'Ms. P Singh', 20),
('s6', 'Art', 'Mr. K Verma', 15);
```
- ```
INSERT INTO ENROLLMENT VALUES
('S00001', 's1', '2023-08-01'),
('S00001', 's2', '2023-08-05'),
('S00002', 's3', '2023-08-10'),
```

```
('S00003', 's1', '2023-08-12'),  
( 'X00009', 's4', '2023-08-15'),  
( 'S00006', 's1', '2023-08-20'),  
( 'S00007', 's2', '2023-08-22'),  
( 'S00008', 's3', '2023-08-25');
```

- ```
SELECT * FROM STUDENT;
```
- ```
SELECT * FROM SOCIETY;
```
- ```
SELECT * FROM ENROLLMENT;
```
- ```
SELECT DISTINCT s.StudentName  
FROM STUDENT s  
JOIN ENROLLMENT e ON s.Roll_No = e.Roll_No;
```
- ```
SELECT SocName FROM SOCIETY;
```
- ```
SELECT StudentName FROM STUDENT WHERE StudentName LIKE 'A%';
```
- ```
SELECT * FROM STUDENT WHERE Course IN ('computer s', 'chemistry');
```
- ```
SELECT StudentName FROM STUDENT WHERE (Roll_No LIKE 'X%9' OR Roll_No  
LIKE 'Z%9');
```
- ```
SELECT * FROM SOCIETY WHERE TotalSeats > 25; -- N = 25;
```
- ```
UPDATE SOCIETY SET MentorName = 'Dr. New Mentor' WHERE SocID = 's5';
```
- ```
SELECT s.SocName
FROM SOCIETY s
JOIN ENROLLMENT e ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName
HAVING COUNT(e.Roll_No) > 5;
```
- ```
SELECT st.StudentName  
FROM STUDENT st  
JOIN ENROLLMENT e ON st.Roll_No = e.Roll_No  
JOIN SOCIETY s ON e.SID = s.SocID  
WHERE s.SocName = 'NSS'  
ORDER BY st.DOB DESC LIMIT 1;
```
- ```
SELECT s.SocName
FROM SOCIETY s
JOIN ENROLLMENT e ON s.SocID = e.SID
```

```
GROUP BY s.SocID, s.SocName
ORDER BY COUNT(e.Roll_No) DESC LIMIT 1;
```

- ```
SELECT s.SocName
FROM SOCIETY s
LEFT JOIN ENROLLMENT e ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName
ORDER BY COUNT(e.Roll_No) ASC LIMIT 2;
```

- ```
SELECT StudentName
FROM STUDENT
WHERE Roll_No NOT IN (SELECT Roll_No FROM ENROLLMENT);
```

- ```
SELECT st.StudentName
FROM STUDENT st
JOIN ENROLLMENT e ON st.Roll_No = e.Roll_No
GROUP BY st.Roll_No, st.StudentName
HAVING COUNT(e.SID) >= 2;
```

- ```
SELECT SocName
FROM SOCIETY s
JOIN ENROLLMENT e ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName
HAVING COUNT(e.Roll_No) = (
 SELECT MAX(cnt) FROM (
 SELECT COUNT(Roll_No) AS cnt FROM ENROLLMENT GROUP BY SID
)
);
```

- ```
SELECT st.StudentName, s.SocName
FROM STUDENT st
JOIN ENROLLMENT e ON st.Roll_No = e.Roll_No
JOIN SOCIETY s ON e.SID = s.SocID;
```

- ```
SELECT DISTINCT st.StudentName
FROM STUDENT st
JOIN ENROLLMENT e ON st.Roll_No = e.Roll_No
JOIN SOCIETY s ON e.SID = s.SocID
WHERE s.SocName IN ('Debating', 'Dancing', 'Sashakt');
```

- ```
SELECT SocName FROM SOCIETY WHERE MentorName LIKE '%Gupta%';
```

- ```
SELECT s.SocName
FROM SOCIETY s
JOIN ENROLLMENT e ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName, s.TotalSeats
HAVING COUNT(e.Roll_No) = 0.10 * s.TotalSeats;
```

- ```

SELECT s.SocName, s.TotalSeats - COUNT(e.Roll_No) AS VacantSeats
FROM SOCIETY s
LEFT JOIN ENROLLMENT e ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName, s.TotalSeats;

```
- ```

UPDATE SOCIETY SET TotalSeats = CAST(TotalSeats * 1.10 AS INTEGER);

```
- ```

ALTER TABLE ENROLLMENT ADD COLUMN FeesPaid TEXT CHECK(FeesPaid IN
('yes', 'No')) DEFAULT 'No';

```
- ```

UPDATE ENROLLMENT SET DateOfEnrollment = CASE
 WHEN SID = 's1' THEN '2018-01-15'
 WHEN SID = 's2' THEN DATE('now')
 WHEN SID = 's3' THEN '2018-01-02'
 ELSE DateOfEnrollment
END;

```
- ```

CREATE VIEW Society_Enrollment_Count AS
SELECT s.SocName, COUNT(e.Roll_No) AS TotalEnrolled
FROM SOCIETY s
LEFT JOIN ENROLLMENT e ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName;

```
- ```

SELECT st.StudentName
FROM STUDENT st
JOIN ENROLLMENT e ON st.Roll_No = e.Roll_No
GROUP BY st.Roll_No, st.StudentName
HAVING COUNT(DISTINCT e.SID) = (SELECT COUNT(*) FROM SOCIETY);

```
- ```

SELECT COUNT(*) AS SocietyCount FROM (
    SELECT SID FROM ENROLLMENT
    GROUP BY SID HAVING COUNT(Roll_No) > 5
);

```
- ```

ALTER TABLE STUDENT ADD COLUMN Mobile_number VARCHAR(10) DEFAULT
'9999999999';

```
- ```

SELECT COUNT(*) AS TotalStudents
FROM STUDENT
WHERE CAST((julianday('now') - julianday(DOB)) / 365.25 AS INTEGER) >
20;

```
- ```

SELECT DISTINCT st.StudentName
FROM STUDENT st
JOIN ENROLLMENT e ON st.Roll_No = e.Roll_No
WHERE strftime('%Y', st.DOB) = '2001';

```

- ```
SELECT COUNT(*) AS Count FROM (  
    SELECT s.SocID  
    FROM SOCIETY s  
    JOIN ENROLLMENT e ON s.SocID = e.SID  
    WHERE s.SocName LIKE 'S%'  
    GROUP BY s.SocID  
    HAVING COUNT(e.Roll_No) >= 5  
);
```

- ```
SELECT
 s.SocName AS 'Society name',
 s.MentorName AS 'Mentor name',
 s.TotalSeats AS 'Total Capacity',
 COUNT(e.Roll_No) AS 'Total Enrolled',
 s.TotalSeats - COUNT(e.Roll_No) AS 'Unfilled Seats'
FROM SOCIETY s
LEFT JOIN ENROLLMENT e ON s.SocID = e.SID
GROUP BY s.SocID, s.SocName, s.MentorName, s.TotalSeats;
```